

Regularized Kalman filter for brain-computer interfaces using local field potential signals

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ABSTRACT

Background: Brain-computer interfaces (BCIs) seek to establish a direct connection from brain to computer, to use in applications such as motor prosthesis control, control of a cursor on the monitor, and so on. Hence, the accuracy of movement decoding from brain signals in BCIs is crucial. The Kalman filter (KF) is often used in BCI systems to decode neural activity and estimate kinetic and kinematic parameters. To use the KF, the state transition matrix, the observation matrix and the covariance matrices of the process and measurement noises must be known in advance, however, in many applications these matrices are not known. Typically, to estimate these parameters, the ordinary least squares method and the sample covariance matrix estimator are used. Our purpose is to enhance the decoding performance of the KF in BCI systems by improving the estimation of the mentioned parameters.

New Method: Here, we propose the Regularized Kalman Filter (RKF) which implements two fundamental features: 1) Regularizing the regression estimate of the state equation to improve the estimation of the state transition matrix, and 2) Use of shrinkage method to improve the estimation of the unknown measurement noise covariance matrix. We validated the performance of the proposed method using two datasets of local field potentials obtained from motor cortex of a monkey (Estimation of kinematic parameters during hand movement) and three rats (Estimation of the amount of force applied by hand as a kinetic parameter).

Results: The results demonstrate that the proposed method outperforms the conventional KF, the KF with feature selection, the Partial least squares, and the Ridge regression approaches.

1. Introduction

The brain's duty is to analyze information and send commands to organs for control. If any obstacles occur in the way of transmitting the order to an organ, the individual will not be able to control the targeted unit. To do this, features from signals that were obtained from brain by brain-computer interfaces (BCIs) systems, were extracted and decoded to improve disrupted functions (Wolpaw and Wolpaw, 2012; Rao, 2013). We decode kinematic and kinetic parameters from local field potential (LFP) recordings, which are generated from dendritic and synaptic activities of hundreds of neurons (Wolpaw and Wolpaw, 2012). Several studies reported multichannel LFPs as an accurate data for obtaining movement related parameters (Zhuang et al., 2010; Beni et al., 2020; Khorasani et al., 2016; Flint et al., 2012), such as force, velocity, position and direction. Many models have been proposed to describe the neural encoding of hand movements, and many algorithms have been

proposed to decode multi-unit activities, including linear regression (Wu et al., 2002; Wu et al., 2003; Wu and Hatsopoulos, 2008; Gupta and Ashe, 2009), partial least square (PLS) (Beni et al., 2020; Khorasani et al., 2016) and Kalman filter (KF) (Wu et al., 2002; Wu et al., 2003; Wu et al., 2004; Wu and Hatsopoulos, 2008; Wu et al., 2009; Gupta and Ashe, 2009; Zhuang et al., 2010; Malik et al., 2011; Li et al., 2009; Dangi et al., 2011; You et al., 2019; Qi et al., 2019; Beni et al., 2020). Several analyses reported that modeling kinematic movements by KF as a stochastic process has improved the decoding accuracy (Wu et al., 2002; Wu and Hatsopoulos, 2008; Zhuang et al., 2010).

Many studies have been done to improve the performance of KF. In most of these studies, KF algorithm had been developed in such a way that mostly led to the complexity of equations, and thereby simulation of the filter. In most studies that have used KF and its derivatives, the state transition and observation matrices are estimated using the ordinary least-square (OLS) method, and the process and measurement noise

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covariance matrices are estimated using the sample covariance matrix (SCM) method (Wu et al., 2002; Wu et al., 2003; Wu et al., 2004; Wu and Hatsopoulos, 2008; Malik et al., 2011; Dangi et al., 2011; Pascucci et al., 2019; Tomar, 2019; Chen et al., 2019; Beni et al., 2020; Veslin et al., 2019). The successful operation of KF depends much on the suitability of the system's model. However, in reality the model of the system is usually either completely unknown or partially known. The effects of the sub-optimal estimation of the model parameters have been investigated in several studies (Heffes, 1966; Nishimura, 1967; Mehra, 1970; Belanger, 1974; Levinbook and Wong, 2008; Gandhi and Mili, 2010; Furrer and Bengtsson, 2007; Nino-Ruiz and Sandu, 2015; Li et al., 2014; Pagnotta et al., 2019). Some of these studies have focused on adaptation of these parameters (Wu and Hatsopoulos, 2008; Dangi et al., 2011; Qi et al., 2019), and others have used autoregressive (AR) model (Milde et al., 2010; Pascucci et al., 2019), lasso (Pagnotta et al., 2019), or ridge regression (RR) methods (Li et al., 2009; Li et al., 2014) to estimate these parameters. Determination of the covariance of the process and measurement noises is also a big obstacle. This almost impedes the ability to run KF. Various studies have examined this issue (Pascucci et al., 2019; Mehra, 1970; Belanger, 1974). In this study, the measurement noise covariance matrix has high dimensions, therefore estimation of this matrix faces many problems such as being ill-conditioned. One of the most widely used solutions to deal with high-dimensional covariance matrix is the shrinkage estimator (Ledoit and Wolf, 2004; Touloumis, 2015; Schäfer and Strimmer, 2005).

The purpose of this study is to enhance the decoding performance of KF in BCI systems by improving the estimation of the mentioned parameters. Here, we propose the Regularized Kalman Filter (RKF) which implements two fundamental features: 1) Regularizing the state equation to improve the estimation of the state transition matrix, and 2) Use of shrinkage method to improve the estimation of the unknown measurement noise covariance matrix. The performance of the proposed method has been investigated on two datasets of LFPs recorded from motor cortex of a monkey (Estimation of kinematic parameters during hand movement) and three rats (Estimation of the amount of force applied by hand as a kinetic parameter). Finally, the results are compared to four other methods.

2. Methods and Materials

2.1. Decoding kinematic parameters: The position and velocity of the monkey's hand

These neural data were previously recorded and have been described (Flint et al., 2012). Briefly, a rhesus macaque (Monkey C) was trained to perform an eight-target, center-out task while operating a two-link manipulandum, using the right hand. In the center-out task, eight square peripheral targets (0° , 45° , 90° , 135° , 180° , 225° , and 270°) around a 10 cm radius circle and a target at the center have been designed. To begin a trial, the center target was highlighted for a random hold time of 0.5-0.6 s as a cue, then a pseudo-randomly selected peripheral target was highlighted, and the center target disappeared. In order to obtain a juice reward, the monkey had to reach the highlighted target within 1.5 s, and hold the cursor for a random time between 0.2-0.4 s.

A 96-channel silicon microelectrode array was implanted intracortically in a monkey over the proximal arm area of the primary motor (M1) and premotor (PMd) cortices contralateral to the arm used to perform the task. Local field potentials were recorded while the monkey performed the motor tasks. Then, LFPs were processed by band-pass filtering between 0.5 and 500 Hz and sampled at 2 kHz. Then signals were digitally notch filtered at 120, 180, and 240 Hz to remove power line noise. Note: the authors just published 95 channels of the LFPs.

2.1.1. Continuous decoding and feature extraction

To pre-process data for continuous decoding, common noise in LFP

signals was removed using the common average reference (CAR) method. Next, to extract the features from LFP signals, the signals were filtered by the 3rd-order Butterworth band-pass filter in five different frequency bands (0-4, 7-20, 70-115, 130-200 and 200-300 Hz). Acquired time series in each frequency band were rectified and then smoothed, using the 3rd-order Savitzky-Golay filter with 100 ms width. Afterward, the mean was subtracted, and to obtain normalized features, all the training data were divided by the standard deviation. Six-time bins, starting 500 ms before the filter iteration time k ($k - 0.5$, $k - 0.4$, $k - 0.3$, $k - 0.2$, $k - 0.1$, k) were used as features. For each LFP signal 2850 features (95 channels*5 frequency bands*6-time lags) were provided. After collecting 10 min. of data during the center-out task by authors, the position and velocity of the manipulandum were decoded.

2.2. Decoding kinetic parameters: The force of the rat's hand

These neural data were previously recorded and have been described (Khorasani et al., 2016); a brief description is given below. Three male Wistar rats were trained to perform a behavioral task. The rats learned to receive a drop of water as a reward by pressing a key which was connected to a force sensor, by their forelimb. Due to the fixation of the force sensor, while the force is applied, the position and orientation of the forelimb were stable. The force sensor controls a lever, which means by applying force, the lever will rotate. When the applying force reaches a predefined threshold (0.15 N), the lever will stop in 90 degrees angle, and after 1.5 s the reward will be received. No cues for the onset and end of trials were defined, and the rats were free to press the force sensor at any time. Intracortical signals were recorded by a 16-channel micro-wire array that was implanted in the primary motor cortex (M1) area contralateral to each rats' preferred arm (all rats used their right hand). LFP signals were obtained from 0.1-500 Hz band-pass filtering and 1 kHz sampling rate.

2.2.1. Continuous decoding and feature extraction

The decoding force features, are obtained from the following steps. At first the CAR method was applied to LFP signals, then the signals were filtered using the 4th order Butterworth filter in each channel through eight different frequency bands (0-4, 4-8, 8-12, 12-30, 30-60, 60-120, 120-200, 200-400 Hz). Afterward, the signals were rectified and smoothed by the 3rd order Savitzky-Golay filter with 300 ms width. The obtained features were normalized and then in 11-time bins from 1 s before the time k with 100 ms time-steps were used. In the end, 1408 features (16 channels*8 frequency bands*11-time lags) were used for force decoding.

2.3. Kalman filter

Assume the vector $x(k) \in \mathbb{R}^M$ and $y(k) \in \mathbb{R}^N$ define the state and the observation of the discrete-time linear dynamical system, respectively. The model of the system could be expressed with two following equations:

1) A process equation

$$x(k+1) = \varphi(k)x(k) + w(k) \quad (1)$$

Where $\varphi(k)$ is a $M \times M$ state transition matrix and the vector $w(k)$ is a $M \times 1$ zero-mean white process noise which the covariance could be calculated as below:

$$E[w(n)w(k)^T] = \begin{cases} Q(k) & n = k \\ 0 & n \neq k \end{cases} \quad (2)$$

where $E[\cdot]$ and $(\cdot)^T$ denote expectation and transposition, respectively.

2) A measurement equation

$$y(k) = H(k)x(k) + v(k) \quad (3)$$

where $H(k)$ is an $N \times M$ observation matrix and the vector $v(k)$ is a $N \times 1$ zero-mean white measurement noise with this covariance formula:

$$E[v(n)v(k)^T] = \begin{cases} R(k) & n = k \\ 0 & n \neq k \end{cases} \quad (4)$$

The estimation algorithm of the Kalman filter has a recursive form, and start with the initial estimate $\hat{x}(0|0)$ and the initial covariance $P(0|0)$. In practice, φ , H , Q , and R might change with the time step, nevertheless, we assume that they are constant, and the two noises and the initial state are mutually independent. Then we can estimate these parameters from training set of the data. The time update equations given *a priori* state estimate and its covariance matrix at $(k+1)$ th recursion

$$\hat{x}(k+1|k) = \varphi \hat{x}(k|k) \quad (5)$$

$$P(k+1|k) = \varphi P(k|k) \varphi^T + Q \quad (6)$$

where $P(k+1|k) = E[(x(k+1) - \hat{x}(k+1|k))(x(k+1) - \hat{x}(k+1|k))^T]$. The filter gain obtained from updating equations is:

$$K(k+1) = P(k+1|k)H^T(HP(k+1|k)H^T + R)^{-1} \quad (7)$$

The measurement update equations which provide *a posteriori* state estimate and its covariance matrix are:

$$\hat{x}(k+1|k+1) = \hat{x}(k+1|k) + K(k+1)(y(k+1) - H\hat{x}(k+1|k)) \quad (8)$$

$$P(k+1|k+1) = [I_M - K(k+1)H]P(k+1|k) \quad (9)$$

where $P(k+1|k+1) = E[(x(k+1) - \hat{x}(k+1|k+1))(x(k+1) - \hat{x}(k+1|k+1))^T]$ and I_M is the $M \times M$ identity matrix (Kalman, 1960; Haykin, 2001; Haykin, 2005).

2.3.1. Alternative forms for updating the covariance

The purpose of this paper is to investigate the performance of Kalman filter in the cases where the dimension of observations is very large. Thus, the dimension of the measurement noise covariance matrix $R \in \mathbb{R}^{N \times N}$ is very large and according to (7) there is an inverse term should be calculated, that increases the computational costs. Therefore, an alternative form for updating covariance, using matrix inversion lemma to reduce the computational load for these cases was used (Bar-Shalom et al., 2004). Hence, the (9) and (7) could be reformed as follows, respectively,

$$P(k+1|k+1)^{-1} = P(k+1|k)^{-1} + H^T R^{-1} H \quad (10)$$

$$K(k+1) = P(k+1|k+1)H^T R^{-1} \quad (11)$$

A summary of Kalman filter is given in Algorithm 1.

Algorithm 1 Kalman filter

Predict step	$\hat{x}(k+1 k) = \varphi \hat{x}(k k)$
	$P(k+1 k) = \varphi P(k k) \varphi^T + Q$
Update step	$P(k+1 k+1)^{-1} = P(k+1 k)^{-1} + H^T R^{-1} H$
	$K(k+1) = P(k+1 k+1)H^T R^{-1}$
	$\hat{x}(k+1 k+1) = \hat{x}(k+1 k) + K(k+1)(y(k+1) - H\hat{x}(k+1 k))$

2.3.2. Parameter fitting

Typically, the state transition and observation matrices in (1) and (3) are estimated by training data using the ordinary least squares (OLS)

method, which minimizes the sum of squares of residuals as much as possible

$$\varphi = \underset{\varphi}{\operatorname{argmin}} \{ \|X_2 - \varphi X_1\|_2^2 \} \quad (12)$$

$$H = \underset{H}{\operatorname{argmin}} \{ \|Y - HX\|_2^2 \} \quad (13)$$

where $\|\cdot\|_2$ denotes the ℓ^2 -norm of a vector. By solving two aforementioned equations, φ and H are given as follows:

$$\varphi = X_2 X_1^T (X_1 X_1^T)^{-1} \quad (14)$$

$$H = Y X^T (X X^T)^{-1} \quad (15)$$

where $X = [x(1), x(2), \dots, x(L)]$, $X_1 = [x(1), \dots, x(L-1)]$, $X_2 = [x(2), \dots, x(L)]$, $Y = [y(1), y(2), \dots, y(L)]$ and L is the length of the training data. Then, the noise covariance matrices are estimated based on the minimum mean-squared error estimates of φ and H

$$Q = \frac{1}{L-1} (X_2 - \varphi X_1)(X_2 - \varphi X_1)^T \quad (16)$$

$$R = \frac{1}{L} (Y - HX)(Y - HX)^T \quad (17)$$

2.4. Regularized Kalman Filter

State space model (1) and (3) represent the system's dynamics and observations. The most important part of the Kalman filter's algorithm is the proper design of the state space model. There are three types of outliers that affect the model components (Gandhi and Mili, 2010). The first type is the observation outlier which occurs via observation/measurement noise $v(k)$ and effects on observations $y(k)$. The next type of the outlier is the innovation outlier, which occurs via system process noise $w(k)$ and affects the state predictions $x(k+1|k)$. The structural outliers considered as the third type which come from structural errors in $\varphi(k)$ and $H(k)$, affect the state predictions, observations, and the state prediction covariance and updated covariance matrices. The presence of these outliers in the Kalman filter leads to a heavily biased solution or even algorithm divergences.

Being unbiased is the desired property of the OLS estimator. However, when there are many predictors or the predictor variables are highly correlated with each other, it can have a huge variance. The general solution to this is the regularization approach. Since the system is dynamic and the state in current time is correlated to the states in previous times, the parameter matrix was fitted to the training data using a form of Tikhonov regularization called ridge regression in the way of response optimization and overfitting avoidance. Then, (12) becomes in the form below

$$\varphi = \underset{\varphi}{\operatorname{argmin}} \{ \|X_2 - \varphi X_1\|_2^2 + \alpha \|\varphi\|_2^2 \} \quad (18)$$

By solving this equation, the state transition matrix is obtained

$$\varphi = X_2 X_1^T (X_1 X_1^T + \alpha I_M)^{-1} \quad (19)$$

$$\alpha = \frac{\lambda_1}{M} \times \operatorname{tr}\{X_1 X_1^T\} \quad (20)$$

where α parameter is the regularization penalty. We discussed in the next sub-section about how to choose it.

The OLS estimator is a good one if $X_1 X_1^T$, when in the form of a correlation matrix, is nearly a unit matrix. Otherwise, the least square estimates are sensitive to a number of errors (Hoerl and Kennard, 1970). In the ridge regression procedure by creating $\lambda_1 I_M$ term, $X_1 X_1^T$ leads to a unit matrix.

As discussed before, we are interested in studying the system's

response to high-dimensional features, without using feature selection methods. Therefore, the measurement noise covariance matrix will be very large. The shrinkage estimator is well-conditioned and more accurate than the sample covariance matrix (SCM). This approach to recognizing a combination of the SCM and identity matrix, shrinks the sample covariance matrix into the identity matrix:

$$\hat{R} = (1 - \beta)R + \beta I_y \quad (21)$$

where I_y is the $N \times N$ identity matrix and β is the optimal shrinkage intensity that defines how much the covariance matrix will shrink toward the identity matrix. According to this procedure, we extend Eq. 17 to the following form.

$$\hat{R} = R + \beta I_y \quad (22)$$

$$\beta = \frac{\lambda_2}{N} \times \text{tr}\{R\} \quad (23)$$

Typical shrinkage estimators use a convex combination for the SCM and the shrinkage target matrix (identity matrix here). In (22) the two items are added together without scaling R . Therefore, the higher values for the diagonal entries would mean less trust in observations. This would shift the balance of trust in the observations versus the transition model. Note, the ridge regression for the transition model should increase bias, thereby increase the diagonal entries in Q , which may somewhat offset the effects of increased diagonal entries in R . The selection of optimal shrinkage intensity parameter is discussed in the next sub-section, and the rest of equations will be the same as Algorithm 1.

2.5. Reviewing other techniques

We compared our proposed procedure with four other methods. A summary of these four methods is given below.

Kalman filter (KF): The first method that we compared with RKF is the conventional Kalman filter defined in Section 2.

Kalman filter with feature selection (KF fs): The second method is the

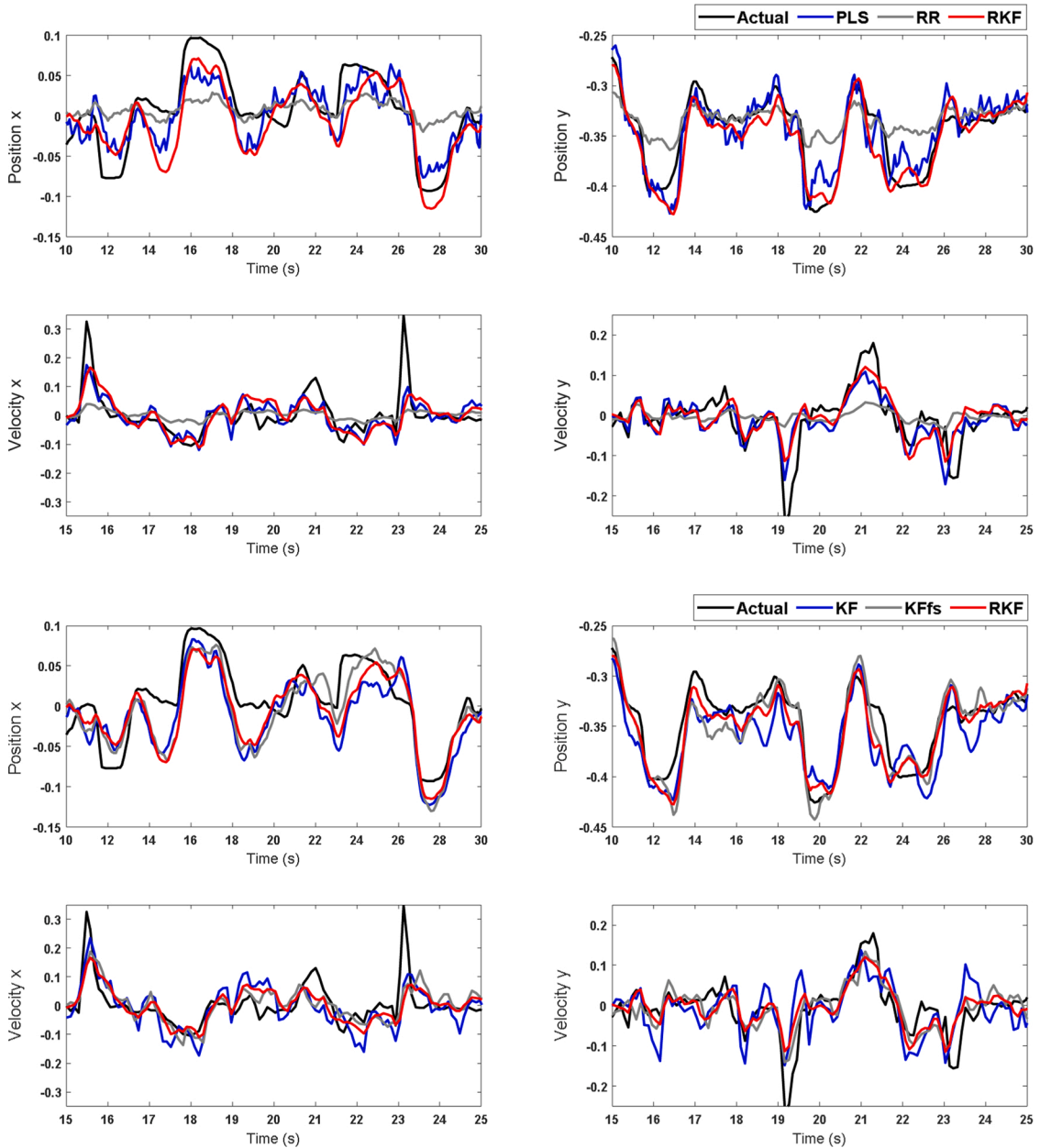


Fig. 1. Sample offline reconstruction trajectories of hand kinematics from the monkey.

Kalman filter with feature selection. The correlation coefficient ($|r|$) between each feature and the recorded signal (position and velocity in dataset 1, and force in dataset 2) was computed and then, the features ranked in descending order based on the absolute values of $|r|$. Then, the Kalman filter was applied on a subset of one feature, on a subset of two features and so forth. Finally, the Kalman filter was applied on all features. The MSE of each response was computed and a subset of features that has minimum MSE was chosen.

Partial Least Square (PLS): The partial least square regression method was compared as the third with RKF which finds a linear regression model ($X = Y\theta + \tau$) by projecting the predicted variables (X , force or position and velocity) and the observable variables (Y , features) to a new space. Bias coefficient was denoted by τ , and θ is the regression coefficient that should be estimated.

Ridge Regression (RR): The last one is the ridge regression method described in Section 2. The kinetic/kinematic signals could be predicted from features, assuming that Y denotes the features and X denotes the kinetic/kinematic signals, therefore, the relevance between variables from RR defined as below:

$$X = GY \quad (24)$$

$$G = \underset{G}{\operatorname{argmin}} \{ \|X - GY\|_2^2 + \gamma \|G\|_2^2 \} \quad (25)$$

$$G = XY^T(YY^T + \gamma I_y)^{-1} \quad (26)$$

$$\gamma = \frac{\lambda_3}{N} \times \operatorname{tr}\{YY^T\} \quad (27)$$

where γ is the ridge regression parameter.

Note, the pre-processing and feature selection for all these methods are the same, and the only difference is in the decoding method.

2.6. Evaluation

To evaluate the performance of the methods, the 10-fold cross-validation method was exploited. In this procedure, the data are partitioned into 10 folds. One of the folds is retained as the validation data to test the model, and the remaining 9 folds are used as training data. Then, this process is repeated 10 times, with each of the 10 folds used exactly once as the validation data. Then, the 10 results from the folds can be

averaged to produce a single estimation.

In order to assess the similarity measure of the real and predicted signals, two criteria have been used; the correlation coefficient (CC) and the normalized mean-square error (NMSE) were computed based on the following equations:

$$CC = \frac{\operatorname{cov}(x, \hat{x})}{\sigma_x \sigma_{\hat{x}}} \quad (28)$$

$$NMSE = \frac{\|x - \hat{x}\|_2^2}{\|x\|_2^2} \quad (29)$$

where x and $\hat{x}$ is the real and predicted signals, respectively. The variance is denoted by σ .

The CC is commonly used for accuracy comparisons in Kalman filtering methods. It shows a statistical relationship between the real and predicted variables. The NMSE is a typical measure of the estimation accuracy of the Kalman filter. It is an estimator of the overall deviations between predicted and measured values. The NMSE generally shows the most striking differences among models.

All of lambdas were chosen from (0, 0.1, 0.3, 0.7, 1, 2 and 10) values by 10-fold cross-validation method from training data.

In this study, to compare the efficiency of the methods, the Wilcoxon sign rank test has been used. This test, which is a nonparametric test, gives us the p-value between the two methods.

3. Results

The proposed method has been tested on two different datasets and compared with four different methods described in the previous section.

The kinematic datasets were recorded over five different experiments. Each of which consisted of 1, 2, 4, 3 and 2 sessions, respectively. In total, there will be 12 datasets in 4 different dimensions (position and velocity in x and y directions). 10-fold cross-validation was applied to datasets. Hand position and velocity were decoded offline, using 10 min. data files. Finally, 120 responses (12 datasets * 10 folds) for each state will be created. Fig. 1 shows examples of position and velocity decoded using each method. In general, the results of KF, KF fs, and RKF methods are more consistent and smoother than the results of PLS and RR methods.

In the kinetic dataset, the LFPs and force of three rats were recorded.

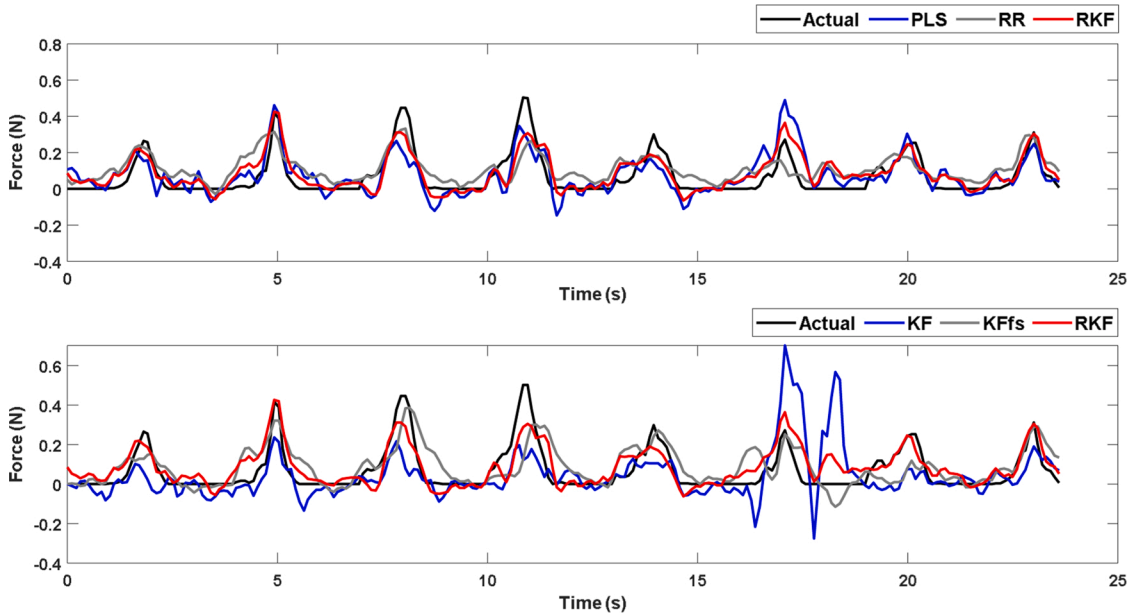


Fig. 2. Sample offline reconstruction trajectories of hand kinetics from the rats.

Table 1

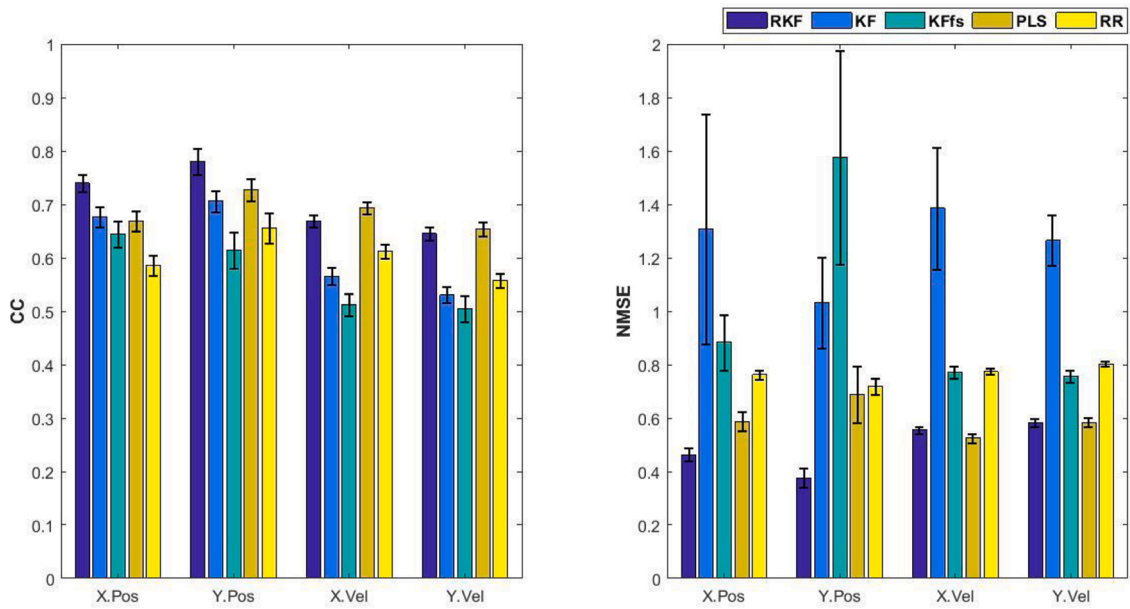
The relative contribution of the two refinements for each animal.

	CC (%)				NMSE			
	Rat 1	Rat 2	Rat 3	Monkey	Rat 1	Rat 2	Rat 3	Monkey
RKF	73.39	69.60	73.05	70.86	0.48	0.60	0.48	0.49
KF	44.81	45.48	50.06	61.95	1.97	2.25	1.31	1.24
KF(R)	69.01	68.05	69.44	71.82	0.70	0.79	0.60	0.54
KF(φ)	44.44	45.23	50.17	61.36	1.86	2.10	1.21	1.13

Table 2

Mean and Standard Error of Correlation Coefficients and of Normalized mean square errors of Kinematic Dataset for Each Method in 120 Repeats.

	CC (%)					NMSE				
	X.Pos	Y.Pos	X.Vel	Y.Vel	Mean	X.Pos	Y.Pos	X.Vel	Y.Vel	Mean
RKF	73.98 $\pm$ 0.84	78.03 $\pm$ 1.21	66.90 $\pm$ 0.56	64.56 $\pm$ 0.62	70.9 $\pm$ 0.003	0.46 $\pm$ 0.01	0.37 $\pm$ 0.01	0.55 $\pm$ 0.005	0.58 $\pm$ 0.005	0.49 $\pm$ 0.005
KF	67.63 $\pm$ 0.92	70.58 $\pm$ 0.96	56.55 $\pm$ 0.78	53.07 $\pm$ 0.77	61.9 $\pm$ 0.001	1.30 $\pm$ 0.21	1.03 $\pm$ 0.08	1.38 $\pm$ 0.11	1.26 $\pm$ 0.04	1.24 $\pm$ 0.07
KF fs	64.43 $\pm$ 1.26	61.41 $\pm$ 1.71	51.21 $\pm$ 1.07	50.47 $\pm$ 1.18	56.9 $\pm$ 0.003	0.88 $\pm$ 0.05	1.57 $\pm$ 0.20	0.77 $\pm$ 0.01	0.75 $\pm$ 0.01	0.99 $\pm$ 0.09
PLS	66.93 $\pm$ 0.93	72.69 $\pm$ 1.05	69.34 $\pm$ 0.60	65.34 $\pm$ 0.65	68.6 $\pm$ 0.002	0.58 $\pm$ 0.01	0.68 $\pm$ 0.05	0.52 $\pm$ 0.005	0.58 $\pm$ 0.01	0.59 $\pm$ 0.02
RR	58.66 $\pm$ 0.94	65.53 $\pm$ 1.39	61.24 $\pm$ 0.65	55.73 $\pm$ 0.66	60.3 $\pm$ 0.003	0.76 $\pm$ 0.01	0.71 $\pm$ 0.01	0.77 $\pm$ 0.005	0.80 $\pm$ 0.005	0.76 $\pm$ 0.004

**Fig. 3.** On the right side of the figure, the mean of the normalized mean squares error and on the left side, the mean of the correlation coefficient of the each method with their standard deviations in all four states (position in x and y direction, velocity in x and y direction) at 120 iterations for the kinematic datasets are shown.

In order to decode the force, the trials were randomly sequenced in each rat, then divided into training and test data using 10-fold cross-validation. Due to the randomness of sequencing the trials, this process was repeated 10 times. In total, there will be 100 responses (10 repeats * 10 folds) for each method and each rat. Fig. 2 demonstrates examples of force decoded using each method. The RKF method, in general, predicts the force more compatible and smoother than the other methods.

Table 1 demonstrates the comparison analysis of performance evaluation on the fundamental improvements. KF(R) is the Kalman filter that used shrinkage estimator for estimating the measurement noise covariance matrix, and KF(φ) is the Kalman filter that used Ridge regression for estimating the state transition matrix. The NMSE columns show that the improvement of the state transition matrix has somewhat reduced the prediction error, and by applying the shrinkage estimator to the measurement noise covariance matrix, this error is significantly reduced. Also, with regard to the CC related columns, we find that improving the measurement noise covariance matrix has a significant effect on the accuracy of filter.

4. Discussion

Our analysis of the neural activity decoded from two datasets demonstrates that the RKF could be used for efficient decoding in neural interface systems. We are interested in studying the system's response to high-dimensional features, without using feature selection methods. We don't want to use feature selection methods, because when the number of variables is large leads to significant computation time. In addition, if all candidate features are equally relevant for the task at hand, removing features will be harmful. Also, there is no guaranty that feature selection will lead to higher performance. Feature selection sometimes provides promising results compared to filters, but it is more time-consuming.

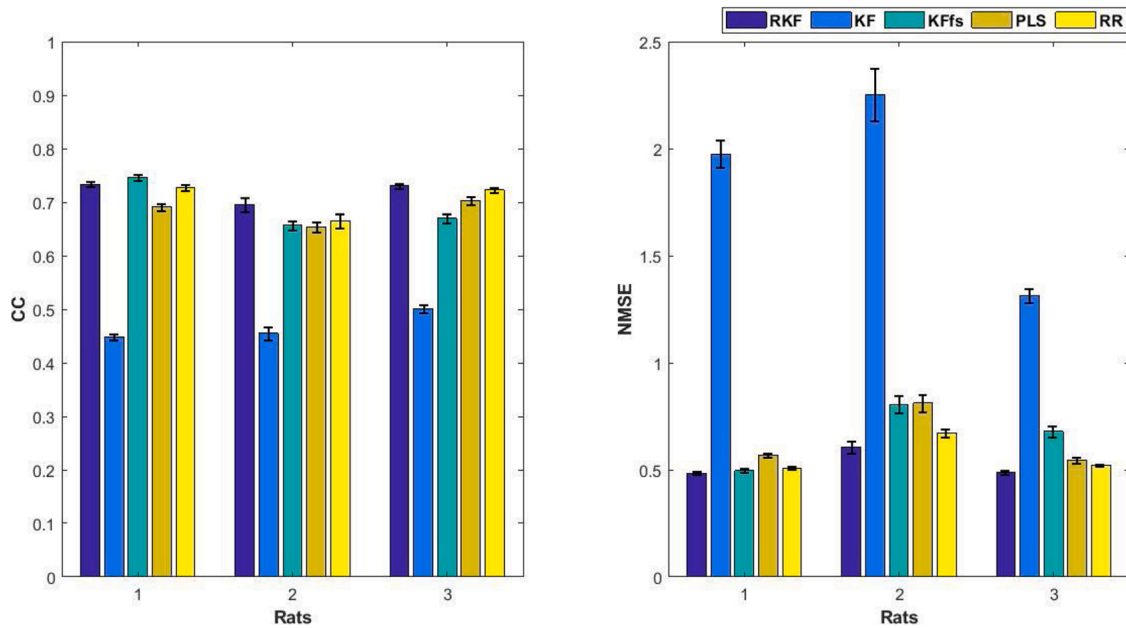
We compare our proposed procedure with four different methods, the conventional Kalman filter, to demonstrate the amount of improvement, the Kalman filter with the simplest type of feature selection, to display the effect of feature selection, As well as the partial least square and the ridge regression methods, as widely used in neural systems decoding.

Table 2 shows the reconstruction accuracy for position and velocity

Table 3

Mean and Standard Error of Correlation Coefficients and Error of Normalized Squares of Kinetic Data Set for Each Method in 100 Repetitions.

	CC (%)				NMSE			
	Rat 1	Rat 2	Rat 3	Mean	Rat 1	Rat 2	Rat 3	Mean
RKF	73.39 ± 0.21	69.60 ± 0.65	73.05 ± 0.25	72.02 ± 0.50	0.48 ± 0.003	0.60 ± 0.01	0.48 ± 0.004	0.52 ± 0.01
KF	44.81 ± 0.25	45.48 ± 0.59	50.06 ± 0.37	46.79 ± 0.51	1.97 ± 0.03	2.25 ± 0.06	1.31 ± 0.01	1.84 ± 0.05
KF fs	74.61 ± 0.27	65.68 ± 0.40	67.01 ± 0.44	69.10 ± 0.50	0.49 ± 0.004	0.80 ± 0.02	0.68 ± 0.01	0.66 ± 0.017
PLS	69.10 ± 0.31	65.41 ± 0.47	70.31 ± 0.36	68.28 ± 0.47	0.56 ± 0.005	0.81 ± 0.02	0.54 ± 0.005	0.64 ± 0.016
RR	72.79 ± 0.27	66.52 ± 0.66	72.27 ± 0.22	70.53 ± 0.53	0.50 ± 0.004	0.67 ± 0.01	0.52 ± 0.002	0.56 ± 0.008

**Fig. 4.** On the right, the mean and standard deviation of the normalized mean square error; and on the left, the mean and the standard deviation of the correlation coefficients of the methods for the kinetic dataset are shown.**Table 4**

p-values for comparing the RKF method with the rest of the methods in the kinetic and kinematic dataset.

RKF vs.	p-Value of CC				p-Value of NMSE			
	Rat 1	Rat 2	Rat 3	Monkey	Rat 1	Rat 2	Rat 3	Monkey
KF	3.54e-18	3.54e-18	3.54e-18	26e-19	3.54e-18	3.54e-18	3.54e-18	0.02e-19
KF fs	0.00888	6.51e-05	3.54e-18	0.02e-19	0.12135	2.84e-06	1.84e-17	0.02e-19
PLS	1.36e-15	1.76e-09	2.20e-09	10e-19	7.89e-13	1.58e-09	2.01e-10	0.70e-19
RR	2.88e-05	1.10e-09	0.000263	0.02e-19	2.70e-08	0.000123	2.70e-08	0.02e-19

of kinematics dataset, measured in NMSE and CC. Each entry is the mean $\pm$ standard error (SE) of NMSE or CC across all sessions. According to the results listed in Table 2, there is a significant difference between CC and NMSE in the proposed method and other methods in predicting the position states (the position in the x direction and in the y direction). As expected, in the case of feature dimensions being too large, the Kalman filter does not respond well, and even feature selection does not outperform the proposed method, despite the improved results. The results of RKF method for velocity states in both directions are close to PLS method, but considering the sum of states, RKF method has better results than the other methods in all four states. These averages are presented in the Fig. 3, along with their standard deviations. It can be seen that the accuracy variation (standard deviation) in the KF and KF fs methods is particularly severe in the NMSE criterion. The RR method compares to the RKF method does not respond well despite a low standard deviation.

The reconstruction accuracy measured in NMSE and CC for force of kinematic datasets have been shown in Table 3. Each entry is the mean $\pm$ SE of NMSE or CC for each rat. As can be seen, as expected from the KF

method, it does not provide a good result in the case of very large feature dimensions. In general, RKF method is more correlated and generates less error than PLS method. For the first rat, the correlation and error of the KF fs and RR methods are close to the RKF method, but the results of the second rat indicate the superiority of the RKF method. Although the correlation of these two methods in the third rat is close to our proposed method, the RKF method is still superior to them due to their calculated error. Based on the last columns of each criterion, which is the mean and standard deviation of the results on the responses of all three rats, one can observe the superiority of the RKF method over the other methods. Fig. 4 shows the mean and standard deviation of CC and NMSE for each rat. As can be seen in the figure, the Kalman filter does not give a good result in terms of large properties, it has both a low correlation and considerable error. The effect of feature dimensions can be seen in KF fs, which decreases the dimension of features giving a better result than the conventional Kalman filter. The results of PLS and RR are also relatively favorable, but as we can see, they are more error and standard deviation than RKF. The RKF method, without any feature selection, yields better results both in terms of correlation and error, and the rate of variation of

its results (standard deviation) is acceptable in different iterations.

The results of the significance testing are presented in Table 4 for the kinematic and kinetic datasets. As shown in the table, there is a significant difference between the RKF method and the other methods (p -value < 0.05).

5. Conclusion

In this study, we implement two refinements to Kalman filter for decoding neural activities and estimate kinetic and kinematic parameters in BCIs systems; firstly, Tikhonov regularization is used to improve estimating the unknown state transition matrix, and later shrinkage estimator is used to improve estimating the unknown observation model noise covariance matrix. We validated the performance of this proposed method using two datasets of local field potentials obtained from motor cortex of a monkey and three rats. Also, we compared our proposed method with conventional Kalman filter, Kalman filter with feature selection, partial least squares (PLS), and Ridge regression methods. The results demonstrate that the proposed procedure significantly outperformed these four methods.

CRediT authorship contribution statement

Matin Asgharpour: Conceptualization, Methodology, Software, Writing - original draft, Writing - review & editing. **Reza Foodeh:** Conceptualization, Methodology, Writing - review & editing. **Mohammad Reza Daliri:** Conceptualization, Methodology, Resources, Supervision, Writing - review & editing.

Declaration of Competing Interest

All authors declared that they have no conflict of interests.

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